

College Football Games and Crimes

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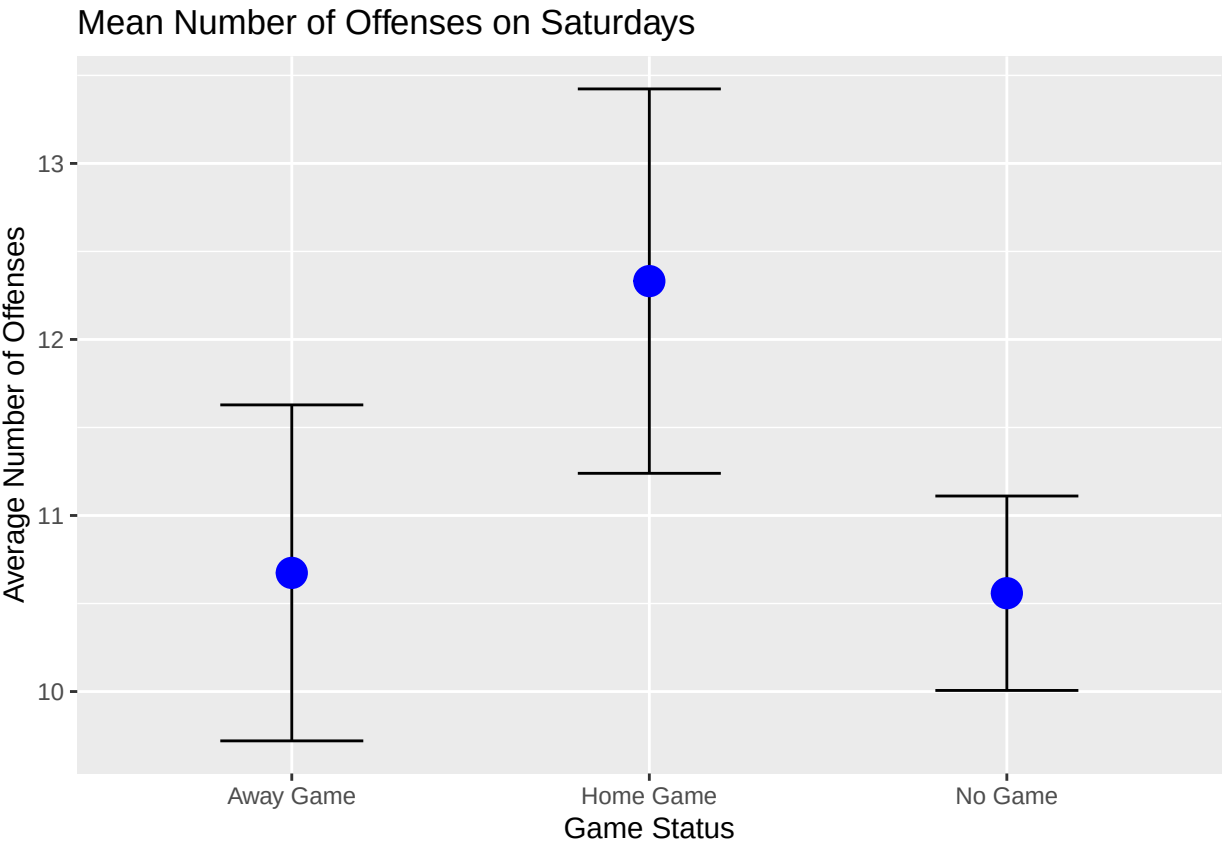
Contents

Table 2: Distribution of Game Days by Day of the Week	1
Distribution of Offenses on Saturdays	2
Distribution of Offenses on Saturdays By Team	3
Appendix: Descriptive Statistics for Count Variables	4
Table 3: College Football Games and Number of Offenses	5
Linear Model	5
Negative Binomial Model	6
Conclusion	7
Table 4: Winning Versus Losing	8
Linear Model	8
Negative Binomial Model	9
Conclusion	10

Table 2: Distribution of Game Days by Day of the Week

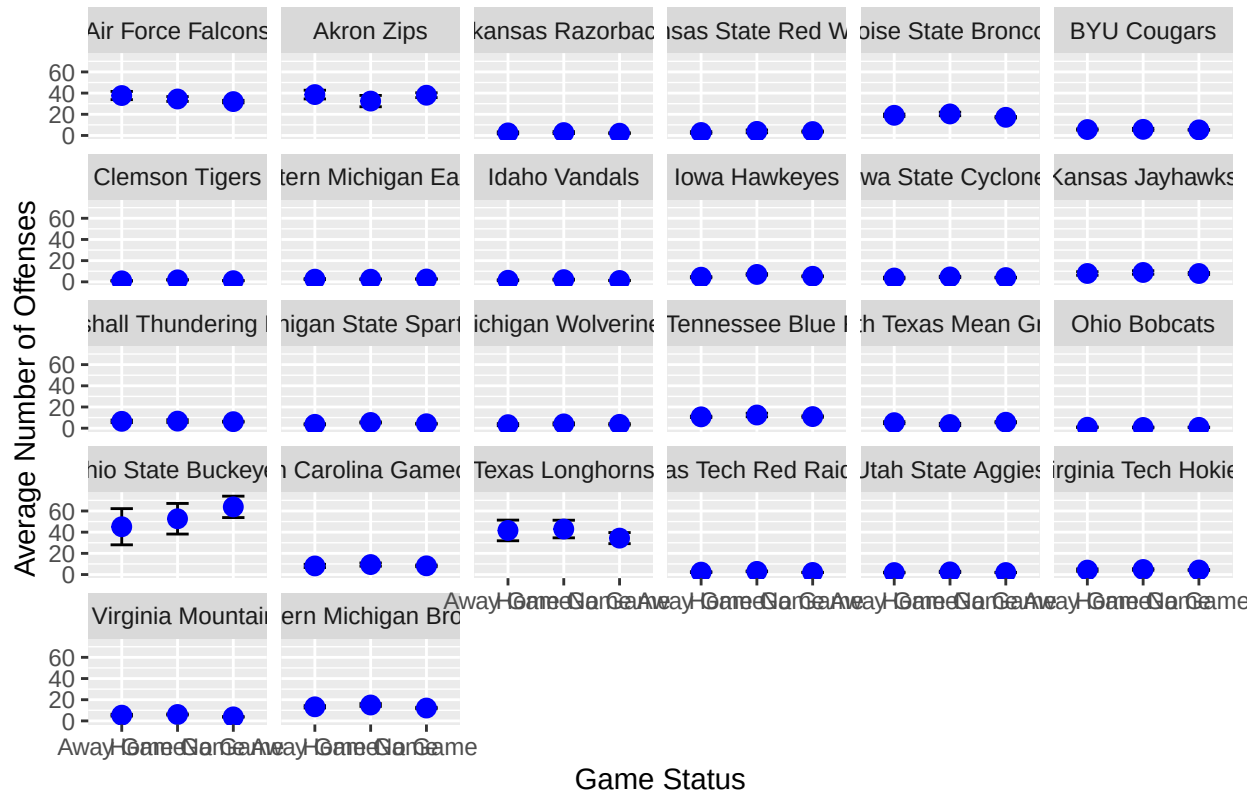
Day of Week	Games	Observations
Sun	7	2522
Mon	2	2522
Tue	1	2522
Wed	3	2522
Thu	51	2496
Fri	17	2522
Sat	866	2522
Total	947	17628

Distribution of Offenses on Saturdays



Distribution of Offenses on Saturdays By Team

Mean Number of Offenses on Saturdays Per Team



Appendix: Descriptive Statistics for Count Variables

offense	Stats	All Days	No Game	Home Game	Away Game
assault	mean	4.536589	4.466812	5.958188	5.219316
assault	std	9.639039	9.506547	12.352515	10.298087
assault	Q1	0.000000	0.000000	0.000000	0.000000
assault	Q2	1.000000	1.000000	2.000000	1.000000
assault	Q3	5.000000	5.000000	6.000000	5.000000
assault	Q4	11.000000	11.000000	14.000000	13.000000
vandalism	mean	4.466757	4.395060	6.000000	5.084507
vandalism	std	10.818816	10.764745	12.541450	10.330082
vandalism	Q1	0.000000	0.000000	0.000000	0.000000
vandalism	Q2	1.000000	1.000000	2.000000	2.000000
vandalism	Q3	4.000000	4.000000	6.000000	5.000000
vandalism	Q4	11.000000	11.000000	16.000000	14.000000

Table 3: College Football Games and Number of Offenses

	Assaults	Vandalism
Home game	0.330 (0.357)	0.246 (0.417)
Away game	-0.167 (0.377)	-0.259 (0.440)
Observations	17 628	17 628
Agency fixed effects	Yes	Yes
Groups	26	26

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3: College Football Games and Number of Offenses
Linear Model

Table 4: College Football Games and Number of Offenses

	Assaults	Vandalism
Home game	0.108+ (0.056)	0.121* (0.057)
Away game	-0.076 (0.061)	-0.036 (0.061)
Observations	17 628	17 628
Log likelihood	-35 114.97	-35 554.16
Agency fixed effects	Yes	Yes
Groups	26	26

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Negative Binomial Model

Conclusion

Comparing our linear regression results to the original table from the paper, neither the home-game nor away-game coefficients were statistically significant predictors of assaults or vandalism in the linear models. In contrast, the original paper reported that home games were associated with approximately a 9% increase in assaults and an 18% increase in vandalism, both with p-values below 0.01.

To further investigate this discrepancy, we fit a negative binomial model, following the methodology described in the original paper. The results showed that home games were associated with an estimated 11.3% increase in assaults and a 12.9% increase in vandalism. However, only the vandalism effect was statistically significant at the $\alpha = 0.05$ level ($p = 0.034$), while the assault effect was borderline significant ($p = 0.055$). Away games were not statistically significant predictors in either model. Overall, these findings partially support the paper's conclusion that home games are associated with increased offense rates, although we were unable to fully replicate the magnitude and statistical significance of the original results.

Table 5: Winning Versus Losing

	Assaults	Vandalism
Home game win	0.320 (0.415)	0.049 (0.484)
Home game loss	0.359 (0.569)	0.672 (0.664)
Away game win	-0.462 (0.556)	-0.735 (0.649)
Away game loss	0.043 (0.463)	0.078 (0.541)
Observations	17 628	17 628
Agency fixed effects	Yes	Yes
Groups	26	26

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: Winning Versus Losing

Linear Model

Table 6: Winning Versus Losing

	Assaults	Vandalism
Home game win	0.114+ (0.064)	0.100 (0.066)
Home game loss	0.095 (0.091)	0.167+ (0.092)
Away game win	-0.068 (0.087)	-0.096 (0.090)
Away game loss	-0.078 (0.076)	0.006 (0.076)
Observations	17 628	17 628
Log likelihood	-35 115.02	-35 553.55
Agency fixed effects	Yes	Yes
Groups	26	26

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Negative Binomial Model

Conclusion

Comparing our linear regression results to the original paper, none of the home-game or away-game outcome variables were statistically significant predictors of assaults or vandalism. In contrast, the original study reported significant effects for both home-game wins and home-game losses.

To further investigate this discrepancy, we fit negative binomial models that matched the methodology used in the original paper. Although some p-values decreased in the negative binomial models, none reached statistical significance at the $\alpha = 0.05$ level. Therefore, we were unable to replicate the paper's finding that home-game outcomes significantly affect assault and vandalism rates. Consistent with the original paper, away-game outcomes did not appear to have a statistically significant effect.